



ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
Guwahati
Course Structure and Syllabus

(From Academic Session 2024-25 onwards)

B. TECH
ELECTRONICS AND TELECOMMUNICATION ENGINEERING
3rd SEMESTER



ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY

Course Structure (From Academic Session 2024-25 onwards)

B. Tech 3rd Semester: Electronics and Telecommunication Engineering

Semester III

Sl. No.	Course Code	Course	Hours per Week			Credit	Marks	
			L	T	P	C	CE	ESE
Theory								
1	ET241301	Electronics Devices	3	0	0	3	30	70
2	ET241302	Digital System Design	3	0	0	3	30	70
3	ET241303	Network Theory	3	0	0	3	30	70
4	ET241304	Signals and Systems	3	1	0	4	30	70
5	MA241305B	Mathematics III-B	3	0	0	3	30	70
6	HS241306	Indian Knowledge Systems	2	0	0	2	-	50
7	AC241307	Language other than MIL	1	1	0	0 (PP/NP)	-	50
Practical								
1	ET241311	Electronics Devices Lab	0	0	2	1	15	35
2	ET241312	Digital System Design Lab	0	0	2	1	15	35
3	ET241313	Network Theory Lab	0	0	2	1	15	35
4	SI241321	Social Internship	0	0	4	2	-	50
TOTAL			18	2	10	23	195	605
Total Contact Hours per week: 30								
Total Credits: 23								

N.B. AC241307 is a Mandatory Audit Course (No Credit). It will be evaluated as PP (Pass) or NP (Not Pass)

DETAILED SYLLABUS

Course Code	Course Title	Hours per week L-T-P	Credit C
ET241301	Electronic Devices	3-0-0	3

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Explain basic principles of semiconductor materials and apply those to electronic devices
- **CO2:** Analyse the characteristics of various electronic devices like diodes, BJTs, FETs, etc.
- **CO3:** Analyse the various circuit configurations of BJTs and FETs.
- **CO4:** Figure out the basic processes required for fabrication of electronic devices.

MODULE 1: Introduction to Semiconductor Physics: (6 Hours)

Review of Quantum Mechanics, electrons in periodic lattices, E-k diagrams. Energy bands in intrinsic and extrinsic silicon, carrier transport, diffusion current, drift current, mobility and resistivity. Sheet resistance, design of resistors. Generation and recombination of carriers, Poisson and continuity equation.

MODULE 2: P-N Junction: (10 Hours)

P-N Junction:

The open circuited P-N junction, space charge region, the biased P-N junction, I-V characteristics. Diode resistance, capacitance, switching time, small signal switching models. Avalanche breakdown, Zener breakdown.

Zener diode, Schottky diode, Varactor diode.

Applications:

Half-wave and full-wave single-phase rectifiers, ripple filtering, peak inverse voltage. Voltage multiplier circuits, clipping and clamping circuits. Voltage regulator circuit using Zener diode. LED, photodiode and solar cell.

MODULE 3: Bipolar Junction Transistor (BJT): (10 Hours)

NPN and PNP transistors, BJT configurations (common-base, common-emitter, common collector) and their input & output characteristics, different modes of operation. Avalanche breakdown & Punch through in BJT. Ebers-Moll model.

BJT biasing:

Operating point, DC & AC load lines, different biasing schemes, stabilization of operating point: thermal runaway and thermal stability. BJT as a switch and amplifier.

MODULE 4: Field Effect Transistor (FET): (10 Hours)

Structure and operation of JFET, the drain and transfer characteristics, biasing of FETs.

MOSFET: Structures and operations of Depletion and Enhancement types of MOSFET drain and transfer characteristics, MOS capacitor, C-V characteristics, Introduction to CMOS.

MODULE 5: Integrated circuit fabrication process: (5 Hours)

Oxidation, diffusion, ion implantation, photolithography, etching, chemical vapour deposition, sputtering, twin-tub CMOS process.

MODULE 6: Novel Devices: (4 Hours)

TFET, FinFET, NCFET, Introduction to 2D materials.

Textbooks/References:

1. Solid State Electronic Devices: G. Streetman and S. K. Banerjee, 7th Edition, Pearson, 2014.
2. Semiconductor Physics and Devices: Donald Neamen and Dhrubus Biswas, McGraw-Hill Education.
3. Physics of Semiconductor Devices, S. M. Sze and K. N. Kwok, 3rd Edition, John Wiley & Sons, 2006.

Course Code	Course Title	Hours per week L-T-P	Credit C
ET241302	Digital System Design	3-0-0	3

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Apply the Boolean algebra and K map to minimize logic function.
- **CO2:** Make use of design and analysis principles of digital system design for combinational logic circuits.
- **CO3:** Make use of design and analysis principles of digital system design for sequential logic circuits.
- **CO4:** Identify the logic families and memory devices.

MODULE 1: Number Systems, Codes and Logic Simplification: (12 Hours)

Number systems representations and their conversion, Signed and unsigned binary numbers, binary arithmetic, one's and two's complement arithmetic, BCD, ASCII, EBCDIC, Gray Codes, Excess-3 codes, code conversion.

Review of Boolean algebra and De Morgan's theorem, SOP & POS forms, canonical forms, minimization of Boolean functions, minimization of Boolean functions using Karnaugh maps (up to 6 variables), Prime implicants, logic gates and circuits.

MODULE 2: Combinational Logic Design: (12 Hours)

Synthesis of combinational logic functions, half and full Adders, subtractors, serial and parallel Adders, BCD adder, comparators, multiplexers, demultiplexers, encoder, decoder, driver & multiplexed display, barrel shifter and ALU. Design of code converters.

MODULE 3: Sequential Logic Design: (14 Hours)

S-R, JK, T, D and Master-Slave JK FF, edge triggered FF, latches, ripple and synchronous counters, shift registers, finite state machines, design of synchronous FSM: Moore and Mealy machine, Algorithmic State Machines charts. Designing synchronous circuits like Pulse Train generator, Pseudo Random Binary Sequence generator, Clock generation.

MODULE 4: Logic Families and Semiconductor Memories: (7 Hours)

TTL NAND gate, specifications, noise margin, propagation delay, fan-in, fan-out, Tristate TTL, ECL, CMOS families and their interfacing, memory elements, concept of Programmable Logic Devices like FPGA. Logic implementation using Programmable Devices.

Textbooks/References:

1. Digital Logic Design and Computer Design: M. Marris Mano, 17th Edition, Pearson Education, 2022.
2. Modern Digital Electronics: R. P. Jain, 4th Edition, Tata McGraw Hill Education, 2010.
3. Digital Electronics: R. Anand, 2nd Edition, Khanna Book Publishing Company, 2024.
4. Digital Electronics-An introduction to theory and practice: Gothman, 2nd Edition, PHI, 1982.
5. Digital Circuits and Systems: Douglas V Hall, McGraw Hill, 1989.

Course Code	Course Title	Hours per week L-T-P	Credit C
ET241303	Network Theory	3-0-0	3

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Analyze the circuit using Kirchhoff's law and Network simplification theorems.
- **CO2:** Analyze electrical networks in the Laplace domain and understand concept of network functions and stability.
- **CO3:** Determine the parameters of a two-port network.
- **CO4:** Apply the knowledge of coupled circuits in system design.
- **CO5:** Design passive filters.

MODULE 1: An Overview of Network Theorems: (12 Hours)

Nodal and Mesh Analysis, matrix approach of network containing voltage sources, current sources and reactance, source transformation and duality. Network theorems: Superposition, reciprocity, Thevenin's, Norton's, Maximum Power Transfer, Compensation and Tellegen's theorem as applied to AC circuits. Three phase circuit analysis: three phase balanced and unbalanced circuit and power calculation.

MODULE 2: Laplace Transform for Network Analysis: (9 Hours)

Steady state and transient response of a network from the S-domain circuit, Linear constant co-efficient differential equation and continuous time domain, analysis of RC, RL, and RLC networks with and without initial conditions using Laplace transform.

MODULE 3: Two Port Networks and Network Functions: (10 Hours)

General principle; Z, Y, Hybrid, Transmission and image parameters and equivalent circuits; Interrelationship between various parameters; Series, parallel and cascade connection of networks and parameters; Driving point and transfer Impedance and Admittance function; Pole-Zeros of Network function, Determination of time domain behavior from pole-zero plot.

MODULE 4: Resonance and Coupled Circuits: (5 Hours)

Resonance—series and parallel; Q factor, BW; Analytical procedure for solving coupled circuits; Mutual Inductance, coefficient of coupling, Dot Convention for coupled circuits; Single tuned and double tuned circuits; Effects of over coupling and selectivity curve; Ideal transformer.

MODULE 5: Passive Filter: (4 Hours)

Filter fundamentals, Classification of filters—Low pass, High Pass, Band Pass and Band Reject Filters.

MODULE 6: Network Topology: (5 Hours)

Graph of a network; Concept of tree, co-tree link, chord, forest, co-forest; Planar and non-planar graph; Incidence matrix, tie set matrix, cut set matrix; Fundamental cut set and tie set schedule; Introduction to equation formulation graphically; Proofs of Nodal and Mesh Analysis; Duality of network.

Textbooks/References:

1. Network analysis: Van, Valkenburg, Prentice hall of India, 2000.
2. Circuits and Network: Sudhakar, A., Shyammmohan, S. P., Tata McGraw-Hill New Delhi, 1994.
3. Engineering Circuit Analysis: A William Hayt, 8th Edition, McGraw-Hill Education.

Course Code	Course Title	Hours per week L-T-P	Credit C
ET241304	Signals and Systems	3-1-0	4

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Characterize different types of signals and systems.
- **CO2:** Analyze LTI systems in continuous-time and discrete-time domains.
- **CO3:** Analyze signals and systems in time and frequency domains.
- **CO4:** Apply principles of Sampling Theorem.

MODULE 1: Introduction to Signals & Systems: (10 Hours)

Signals and systems as seen in everyday life and in various branches of science/ engineering.

Signals: Definition and representation, Classifications of signals. Some special signals of importance: unit impulse, unit step, sinusoid, complex exponential. Continuous and discrete-time signals.

Signal properties: periodicity, causality, absolute integrability, deterministic and stochastic.

Signal operations: time scaling, time shifting, time reversal, amplitude scaling, amplitude reversal, addition and multiplication of signals.

Systems: Definition and representation, Classifications of systems. Continuous and discrete-time systems.

System properties: linearity: additivity and homogeneity, time-invariance, causality, stability, invertibility, realizability.

MODULE 2: Continuous time and Discrete time LTI systems: (8 Hours)

Linear time-invariant (LTI) system, impulse response and step response, convolution integral and convolution sum, input-output behavior with aperiodic convergent inputs. Characterization of causality and stability of LTI systems. System representation through differential equations and difference equations.

MODULE 3: Fourier Analysis: (10 Hours)

The notion of a frequency response and its relation to the impulse response, Fourier series representation, Fourier Transform, convolution/multiplication and their effect in the frequency domain, magnitude and phase response, Fourier domain duality, Discrete-Time Fourier Transform (DTFT), Parseval's Theorem.

MODULE 4: Laplace Transform: (6 Hours)

Definition and properties, notion of Eigen functions of LTI systems, a basis of Eigen functions, region of convergence. System function, poles and zeros of system functions and signals. Laplace domain analysis for continuous time signals and systems.

MODULE 5: The Z-Transform: (6 Hours)

Definition and properties, Eigen functions, region of convergence. System function, poles and zeros of systems and sequences. Z-domain analysis for discrete-time signals and systems.

MODULE 6: Sampling and Reconstruction: (5 Hours)

Sampling Theorem and its implications, spectra of sampled signals. Reconstruction: ideal interpolator, zero-order hold and first-order hold. Aliasing and its effects.

Textbooks/References:

1. Signals and Systems: A.V. Oppenheim, A.S. Willsky and I.T. Young, Prentice Hall, 1983.
2. Signals and Systems - Continuous and Discrete: R.F. Ziemer, W.H. Tranter and D.R. Fannin, 4th edition, Prentice Hall, 1998.
3. Signals and Systems: R. Anand, Khanna Publishing House, 2019.
4. Circuits and Systems: A Modern Approach: Papoulis, HRW, 1980.
5. Signal Processing and Linear Systems: B.P. Lathi, Oxford University Press, 1998.

6. Introduction to Signals and Systems, Douglas K. Lindner, Mc-Graw Hill International Edition: 1999.
7. Signals and Systems, Simon Haykin, Barry van Veen, John Wiley and Sons (Asia) Private Limited, 1998.

Course Code	Course Title	Hours per week L-T-P	Credit C
MA241305B	Mathematics III-B	3-0-0	3

Course Outcomes: At the end of the course the students will be able to

- **CO1:** Apply concepts of vector spaces, linear transformations, and inner product spaces to analyse linear systems and perform orthogonalization in engineering and scientific computations.
- **CO2:** Apply fundamental probability concepts, including conditional probability, independence, and Bayes' Theorem, to model and analyse random events in engineering contexts.
- **CO3:** Analyse discrete and continuous random variables using probability distributions and joint distributions, and apply these concepts to model uncertainties in real-world systems.
- **CO4:** Compute expectation, variance, moments, and covariance to quantify and interpret the statistical behaviour of random variables in practical scenarios.
- **CO5:** Model and analyse random processes by evaluating stationarity, ergodicity, and correlation functions to address uncertainties in engineering systems.

MODULE 1: Linear Algebra:

(12 Hours)

Vector space: Definition and properties of vector spaces, definition and properties of subspace, linear independence, bases and dimension.

Linear transformation: Definition, linear maps and matrices, rank-nullity theorem, Eigen values and Eigen vectors.

Inner product space: Definition, Euclidean and unitary space, Schwartz's inequality, orthogonality, norms, orthonormal bases, Gram-Schmidt orthogonalization process.

MODULE 2: Probability:

(5 Hours)

Classical and axiomatic definitions, sample space, joint probability, conditional probability, Bayes' Theorem, independent events.

MODULE 3: Random variables:

(12 Hours)

Definition, continuous and discrete random variables, cumulative distribution function (CDF), probability functions (PMF & PDF), Probability Distributions: Gaussian, uniform and exponential, binomial distribution, Poisson, joint distributions (JPMF & JPDF), central limit theorem (Statement only).

MODULE 4: Operation on one random variable:

(6 Hours)

Expectation, variance and standard deviation, Covariance, Raw moment & central moments.

MODULE 5: Random Process:

(10 Hours)

The random process concept, stationarity, wide-sense stationarity, ergodicity, auto-correlation and cross-correlation functions, auto-covariance and cross-covariance functions, examples of random processes.

Textbooks/References:

1. Introduction to Linear Algebra: Strang G., Wellesley-Cambridge Press; 4th Edition, 2009
2. Linear Algebra and its Application: David C Lay, Pearson.
3. Probability, Random Variables and Random Signal Principles, Peyton Z Peebles Jr. , McGraw-Hill, 4th Edition, 2017.

4. Probability and Random Processes with Applications to Signal Processing: Henry Stark and John W. Woods, Pearson, 3rd Edition, 2012.
5. Advanced Engineering Mathematics: Erwin Kreyszig, John Wiley & Sons, 9th Edition, 2006.
6. Introduction to Probability Theory: P. G. Hoel, S. C. Port and C. J. Stone, Universal Book Stall, 2003 (Reprint).
7. A text book of Engineering Mathematics: N.P. Bali and Manish Goyal, Laxmi Publications, 2010 (Reprint).
8. Engineering Mathematics (for semester III): Veerarajan T., Tata McGraw-Hill, 2010.
9. Introduction to Probability and Statistics for Engineering and Scientists: S. M. Ross, Academic Press, 3rd Edition, 2004.
10. Probability, Random Variables, and Stochastic Processes: A. Papoulis and S. U. Pillai, Tata McGraw-Hill, 4th Edition, 2002.

Course Code	Course Title	Hours per week L-T-P	Credit C
HS241306	Indian Knowledge Systems	2-0-0	2

Course Objectives:

1. To introduce students to the foundational concepts of Indian Knowledge Systems (IKS).
2. To explore the contributions of ancient and medieval India in science, technology, mathematics, and philosophy.
3. To develop an appreciation for traditional Indian knowledge and its relevance in modern contexts.
4. To encourage critical thinking about indigenous knowledge systems and their applications.

Course Outcome (CO):

After completion of the course, the students will be able to

- **CO1:** Understand the key principles of Indian Knowledge Systems.
- **CO2:** Recognize India's contributions to science, mathematics, medicine, and engineering.
- **CO3:** Analyze the interdisciplinary nature of IKS and its connections with modern technology.
- **CO4:** Appreciate the philosophical and ethical dimensions of traditional Indian knowledge.

MODULE 1: Introduction to Indian Knowledge System (IKS) (4 Lectures)

- Definition, scope, and significance of IKS
- Sources of Indian knowledge: Vedas, Upanishads, Puranas, Smritis, and other texts
- Classification of knowledge in Indian tradition (Vidyas and Kalas)

MODULE 2: Science & Technology in Ancient India (12 Lectures)

- Contributions to mathematics (Zero, Decimal System, Vedic Mathematics, Aryabhata, Brahmagupta)
- Astronomy & Space Science (Surya Siddhanta, Planetary Motion, Concept of Time)
- Metallurgy and material sciences (Iron Pillar, Wootz Steel, Ayurvedic alloys)
- Civil engineering & architecture (Temple architecture, Town planning, Water management)

MODULE 3: Indian Philosophical & Ethical Systems (6 Lectures)

- Key schools of Indian philosophy (Nyaya, Vaisheshika, Samkhya, Yoga, Vedanta)
- Concept of Dharma, Karma, and sustainable living
- Ethics in science and technology from an Indian perspective

MODULE 4: Traditional Medicine & Wellness Systems (4 Lectures)

- Foundations of Ayurveda, Yoga, and Siddha
- Holistic health and wellness practices
- Relevance of traditional medicine in modern healthcare

MODULE 5: IKS in Contemporary Applications (4 Lectures)

- Revival of traditional knowledge in modern science and engineering
- Case studies (e.g., Panchakarma in wellness tech, Vastu in architecture)
- Government initiatives (National Education Policy 2020, IKS Division under AICTE)

Text Books:

1. "Indian Knowledge Systems" – Kapil Kapoor & Avadhesh Kumar Singh
2. "Ancient Indian Leaps into Mathematics" – B.S. Yadav
3. "Science and Technology in Ancient India" – O.P. Jaggi
4. "The Positive Sciences of the Ancient Hindus" – Brajendranath Seal
5. AICTE Model Curriculum for IKS (Available online)

Online Resources:

- NPTEL / SWAYAM courses on IKS
- AICTE-IKS Portal (<https://iksindia.org/>)
- Digital Library of India (<https://dli.gov.in>)

Course Code	Course Title	Hours per week L-T-P	Credit C
AC241307	Language other than MIL	1-1-0	0

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Demonstrate a clear understanding of basic grammar rules including subject-verb agreement, tenses, and cases, and apply them in constructing grammatically correct sentences.
- **CO2:** Identify and correct common grammatical and usage errors in written English, enhancing sentence effectiveness and clarity.
- **CO3:** Enhance their vocabulary through the use of synonyms, antonyms, idiomatic expressions, and phrasal verbs, improving both written and spoken communication.
- **CO4:** Apply verbal reasoning skills to comprehend, arrange, and complete sentences and paragraphs effectively.
- **CO5:** Develop strong reading comprehension skills using skimming, scanning, and intensive reading techniques to extract key information from various texts.
- **CO6:** Produce coherent and concise writing through precis writing and essay composition, demonstrating logical flow, clarity, and appropriate structure.

MODULE 1: Basic grammar usage

(10 Lectures)

Subject-verb agreement, narration, tenses

Case: Subjective case, possessive case, objective case

Writing effective sentences, Spotting errors

MODULE 2: Vocabulary Building

(6 Lectures)

Synonyms and Antonyms, One word substitution,

Idioms and Phrases and Phrasal Verbs

MODULE 3: Verbal ability

(8 Lectures)

Cloze test

Sentence rearrangement, Sentence completion,

Paragraph Jumbles

MODULE 4: Writing and Reading Ability

(8 Lectures)

Reading subskills, Skimming, Scanning and Intensive reading

Reading Comprehension, Precis Writing, Essay Writing

Text Books

1. Objective General English: S.P.Bakshi, Arihant Publication India Ltd., 2021
2. Descriptive English: S.P.Bakshi, Arihant Publication India Ltd., 2016

Course Code	Course Title	Hours per week L-T-P	Credit C
ET241311	Electronic Devices Lab	0-0-2	1

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Test the working of different diodes, transistors, CRO probes and measuring instruments.
- **CO2:** Design the circuits with basic semiconductor devices.
- **CO3:** Measure and record the experimental data, analyze the results, and prepare a formal laboratory report.

LIST OF EXPERIMENTS

1. To study the different components and equipment of Electronics Laboratory.
2. To study V_D - I_D characteristics of Silicon Diode.
3. To study V-I Characteristics of Zener diode and Zener diode as a voltage regulator.
4. To study Half-Wave rectifier with and without filter.
5. To study Full-Wave rectifier with and without filter.
6. To study Bridge rectifier with and without filter.
7. To plot the input and output characteristics of a transistor in CB Configuration.
8. To plot the input and output characteristics of a transistor in CE Configuration, and to find the β value.
9. To study Drain characteristics and Transfer characteristics of JFET.
10. To study characteristics of MOSFET.

Course Code	Course Title	Hours per week L-T-P	Credit C
ET241312	Digital System Design Lab	0-0-2	1

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Simplify, design and implement Boolean expression/ adders /subtractors using basic/universal gates.
- **CO2:** Design and implement the various combinational and sequential circuits.
- **CO3:** Analyze the results, and prepare a formal laboratory report.

LIST OF EXPERIMENTS

1. To familiarize with logic gate IC packages and to verify the truth tables of logic gates and DE Morgan's theorem.
2. To simplify Boolean expression and verify them.
3. To design and implement full-adder and full-subtractor circuits.
4. To design and implement parallel adder using IC7483.
5. To design and implement binary-to-gray code converter circuit.
6. To design and implement MUX and DEMUX using logic gates and MUX/DEMUX ICs.
7. To verify workings of S-R, J-K, Master-slave J-K, D and T flip-flops.
8. To design and implement shift register using flip-flop.
9. To design and implement Mod-N synchronous counter using flip-flop.
10. To design and implement multiplier circuit.

Course Code	Course Title	Hours per week L-T-P	Credit C
ET241313	Network Theory Lab	0-0-2	1

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Test for validity of the network theorems.
- **CO2:** Determine the response of different circuits.
- **CO3:** Measure and record the experimental data, analyze the results, and prepare a formal laboratory report.

LIST OF EXPERIMENTS

1. To verify the principle of superposition with dc and ac sources.
2. To verify Thevenin's, Norton's and Maximum power transfer theorems in ac circuits.
3. To verify Tellegen's theorem for two networks of the same topology.
4. To determine transient response of current in RL and RC circuits with step voltage input.
5. To determine transient response of current in RLC circuit with step voltage input for under damp, critically damp and over damp cases.
6. To determine frequency response of current in RLC circuit with sinusoidal ac input.
7. To determine z and h parameters (dc only) for a network and computation of Y and ABCD parameters
8. To determine driving point and transfer functions of a two-port ladder network and verify with theoretical values.
9. To study frequency response of Passive Filter circuit.
